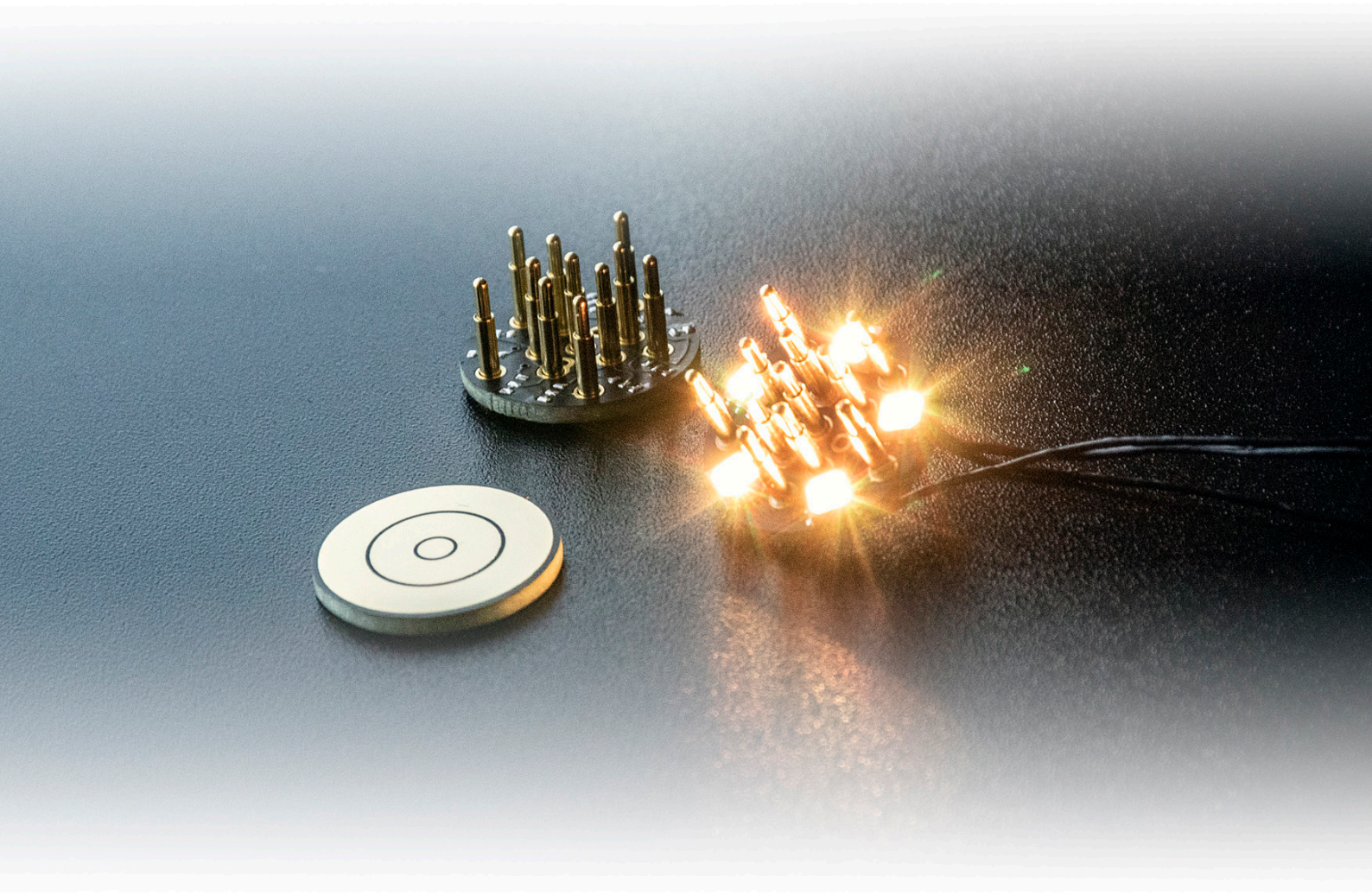
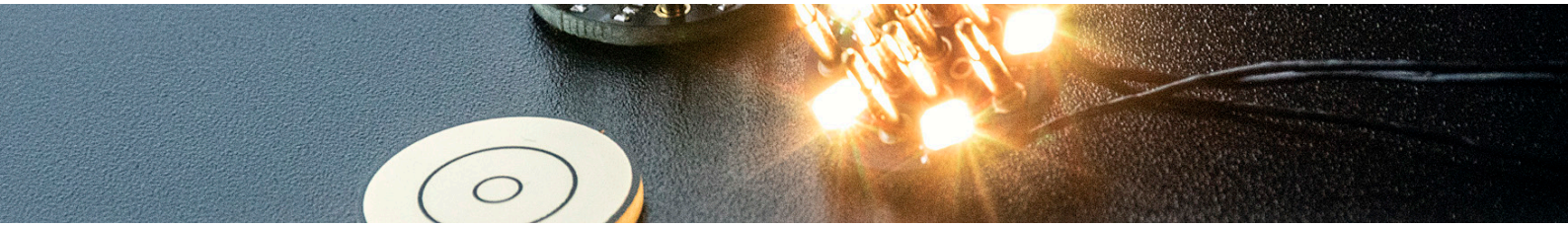
NPXL - Pixelblade Lit Connector



by ShtokCustomWorx

2018



NPXL Pixelblade Lit Connector
by ShtokCustomWorx

User Manual

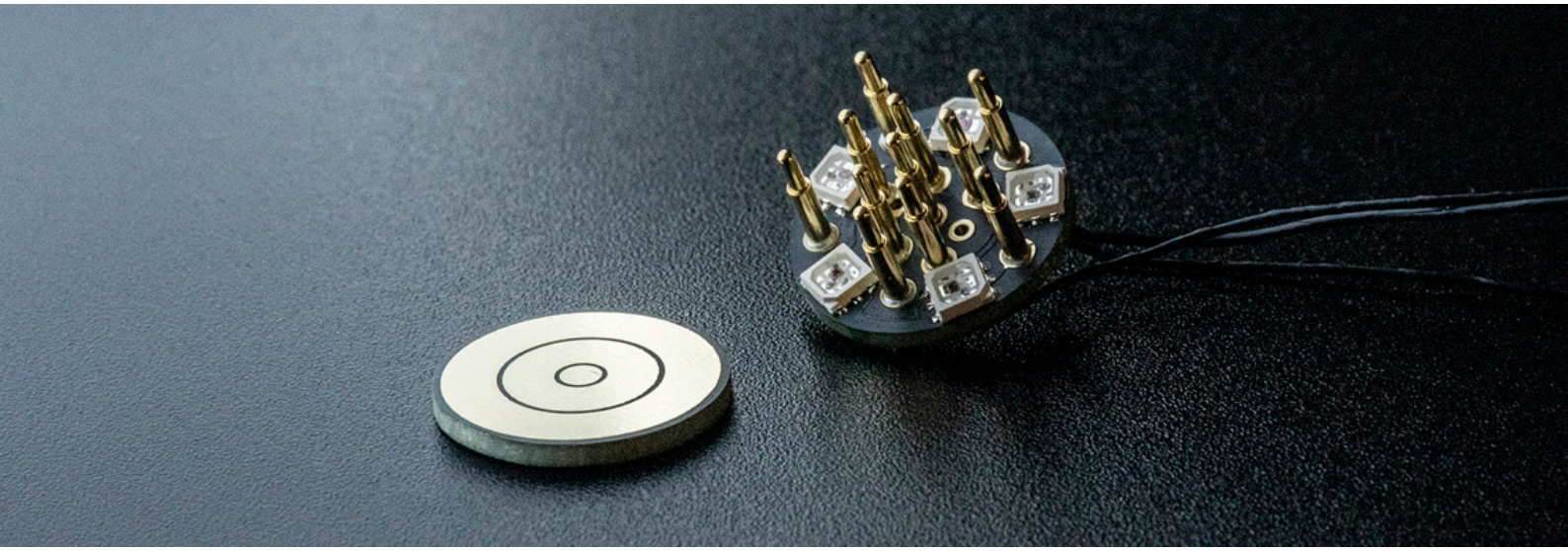
Updated: 11.09.2019

This manual shows and explains how to install and operate SCW NPXL Pogo-pins connectors for Neopixel blade setup.

Contents

Features (P – 2)
1. Assembly variants (P – 3)
2. Data input options (P – 4)
3. Wiring diagrams (P – 5-8)
4. Mounting options (P – 9-10)
3D-printed pcb Holders:
– Hilt pcb Holder
– Blade pcb Holder
5. Tests charts (P – 11-12)
– Current draw test
– Max ratings test
6. Wire gauge recommendations (P – 13)

FEATURES



Specifications and features:

– High quality components, manufacturing and assembling
– Black pcbs with Gold Plated pads
– Compatible wih TCSS/VV/PlecterPixel Blade side pcbs/blades
– Rated for 20 Amperes max (up to 15 Amperes recommended) without current losses
– Reliable electrical connection
– 3-pole contact type: 1-pin Data, 5-pin Positive, 5-pin Negative
– High quality spring loaded Pogo pins (short and long style)
– 5 on-board RGB Neopixel leds with separate Data input for Blade Plugs backlight
– Dimensions:

Hilt pcb: 17.5x17.5x2.6mm

Blade pcb: 17.5x17.5x1.6mm

Where to buy:

– ShtokCustomWorx (Russia)
– Saberbay Etsy store (USA)
– KR-sabers TheSaberArmory store (UK)



Old version (2018-2019)



Version 2 (2019)

Please watch these videos for full features demonstration:

Link to the demonstration video #1
Link to the demonstration video #2

Please watch this video for assembly demonstration:

Link to the assembly video

1 ASSEMBLY VARIANTS

Long pins

for regular/dueling sabers





Short pins

for “Thin-Neck” sabers and chassis

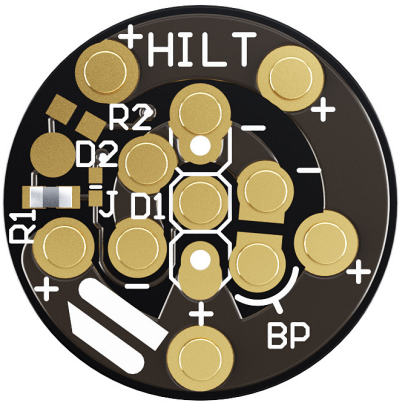




2 DATA INPUT OPTIONS

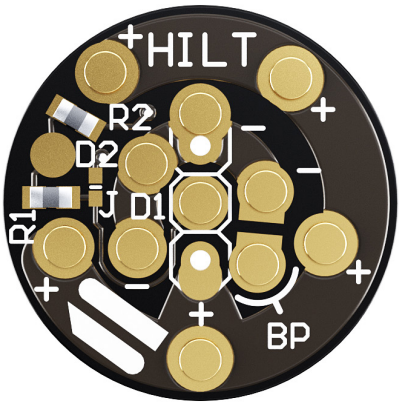
V1 – only blade, no pcb leds

This way Data lead controls only blade, no leds on connector pcb.
Data lead can be wired whether to D1 pad if 330 ohm resistor is on the sound board or to D2 pad with on-board 330 ohm 0603 resistor R1.



V2 – blade + pcb leds in parallel

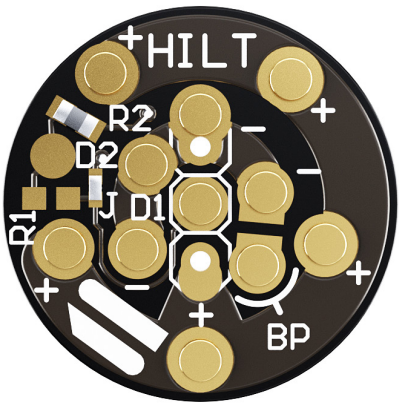
This way a single Data lead controls both blade and on-board connector leds in parallel.
Data lead is wired to D2 pad with on-board 330 ohm 0603 resistors R1 and R2.



V3 – blade + pcb leds in series

This way a single Data lead controls both blade and on-board connector leds in series. Signal goes through 5 pcb leds first, then through jumper J to the blade strips.
Data lead is wired to D2 pad with on-board 330 ohm 0603 resistor R2 and bridged jumper J.

Make sure to add additional 5 leds to the config file.



V4 – blade + pcb leds independent

This way blade and on-board connector leds are controlled separately by independent Data leads. Only for those sound boards that have 2 independent Neopixel Data outputs.
Connector leds Data lead is wired to pad D2 with on-board 330 ohm 0603 resistor R2, blade Data lead is wired to pad D1.

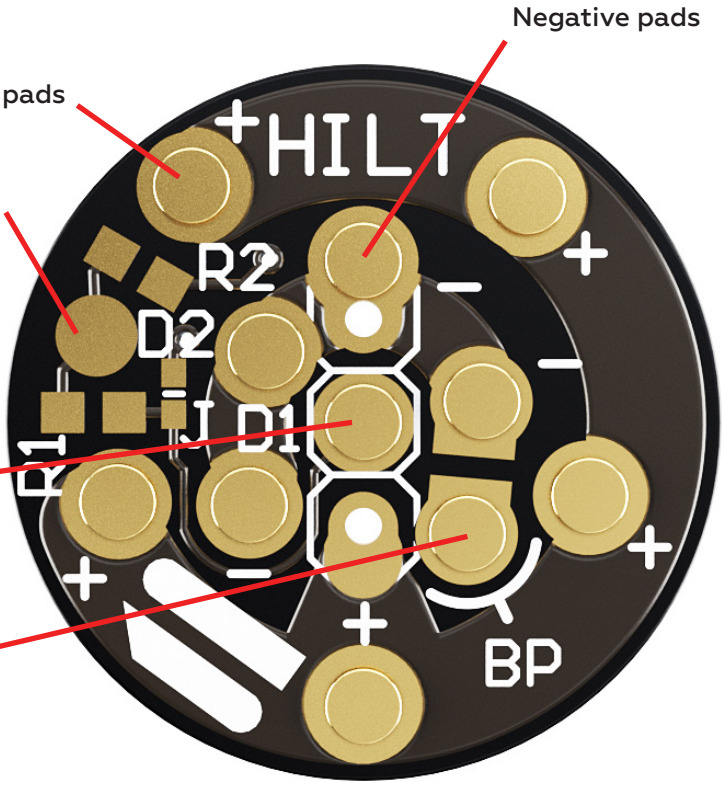


WIRING DIAGRAMS

Hilt pcb

Wire Data lead to D1 or D2 pads, depending on the setup (please follow instructions on page 4). Wire Positive and Negative leads directly to corresponding pins pads.

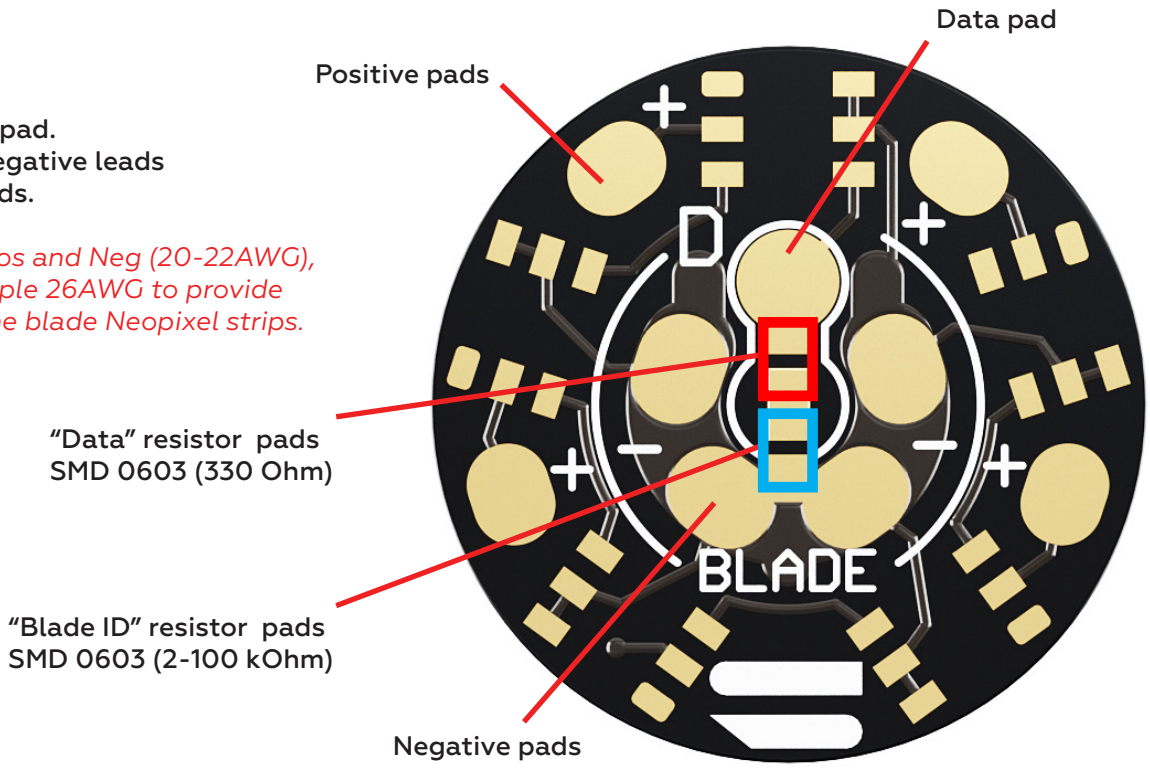
Use thick wires for Pos and Neg (20-22AWG), double 24AWG or triple 26AWG to provide enough current to the blade Neopixel strips.



Blade pcb

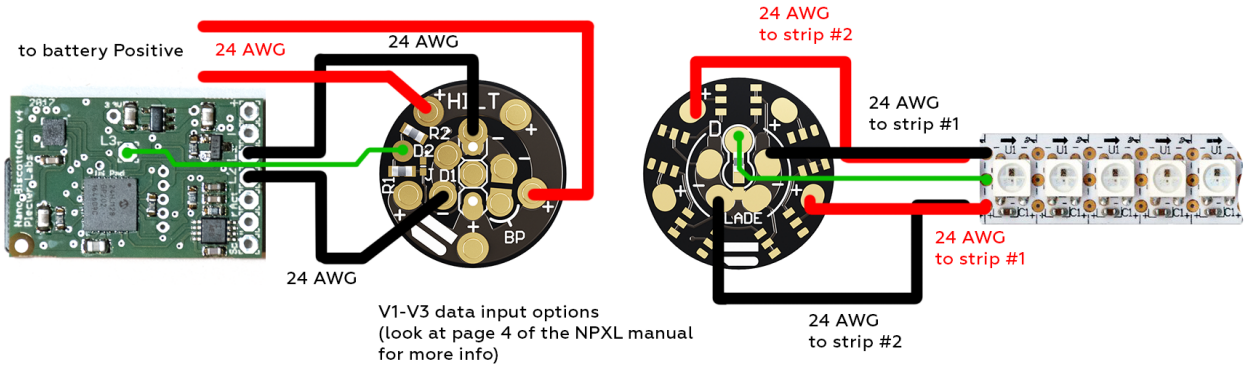
Wire Data lead to D pad.
Wire Positive and Negative leads to corresponding pads.

Use thick wires for Pos and Neg (20-22AWG), double 24AWG or triple 26AWG to provide enough current to the blade Neopixel strips.



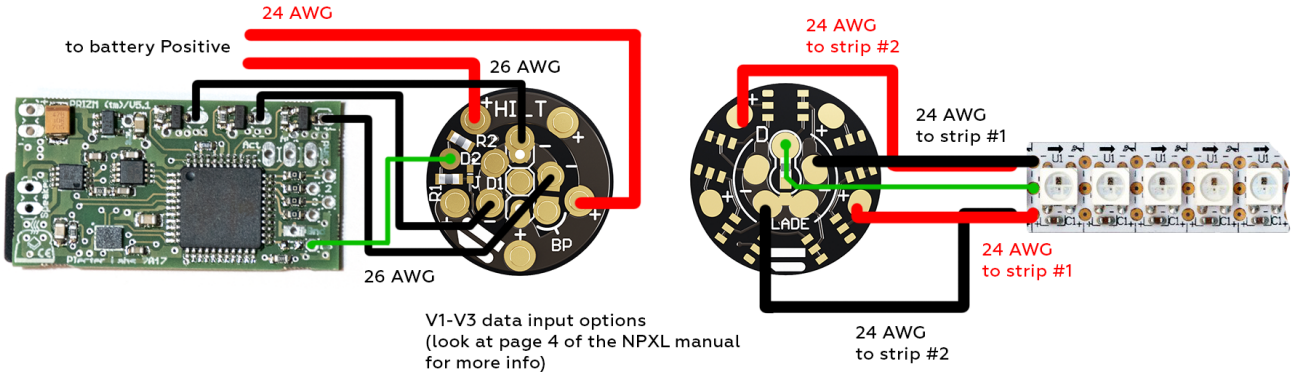
WIRING DIAGRAMS

NANO BISCOTTE



Solder wires to any corresponding pins on the back of Hilt pcb.
Recommended Power wires (Positive and Negative GND) - 22 AWG single or 26 AWG dual.

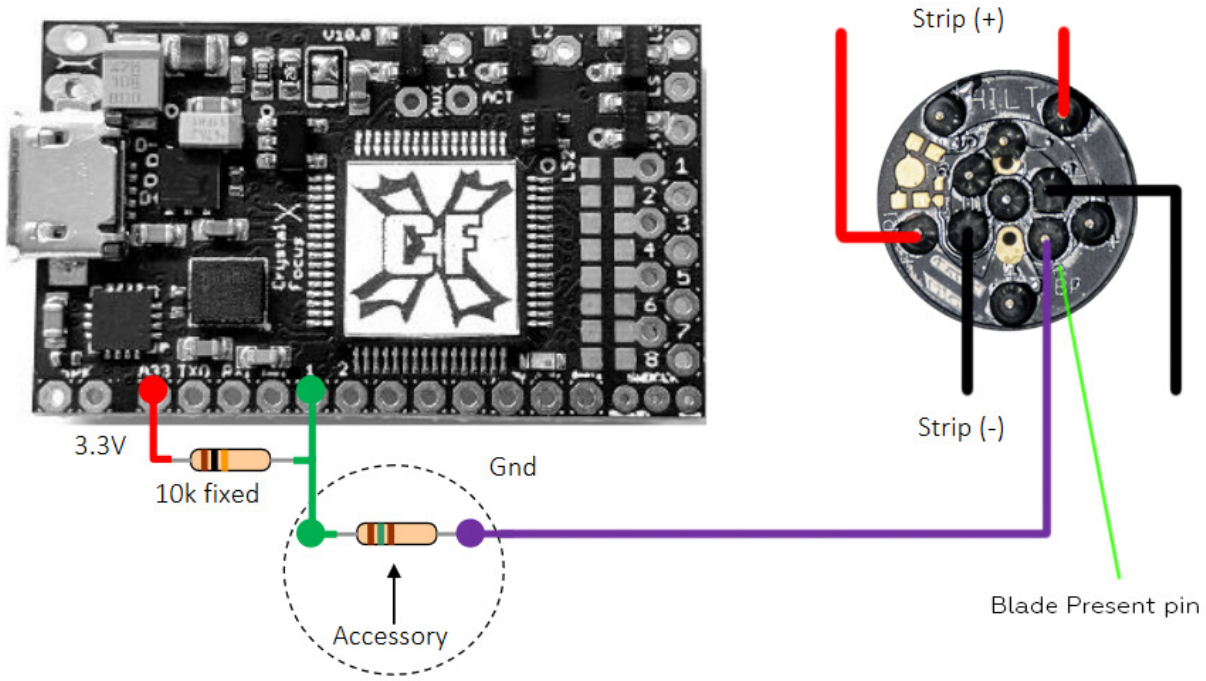
PRIZM v5.5



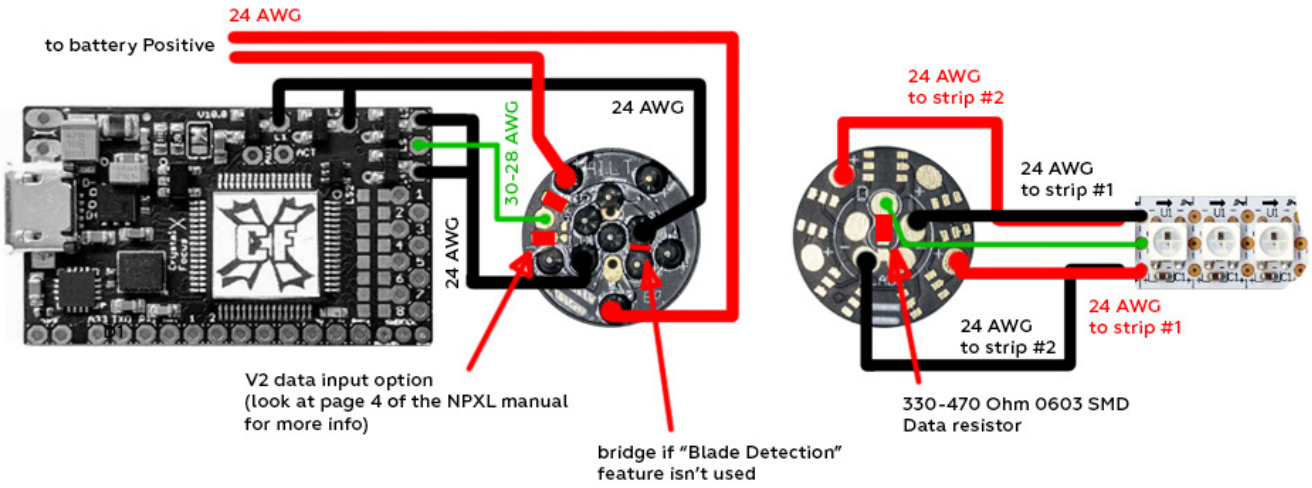
Solder wires to any corresponding pins on the back of Hilt pcb.
Recommended Power wires (Positive and Negative GND) - 22 AWG single or 26 AWG dual.

WIRING DIAGRAMS

CFX Blade Detection wiring with NPXL lit connector by SCW

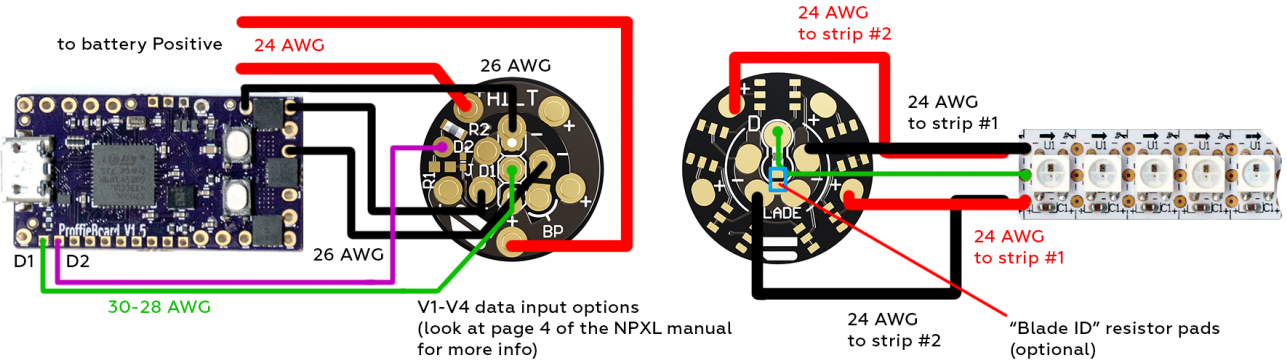


CFX wiring with NPXL lit connector by SCW



WIRING DIAGRAMS

TEENSYSABER/PROFFIEBOARD



Solder wires to any corresponding pins on the back of Hilt pcb.
Recommended Power wires (Positive and Negative GND) - 22 AWG single or 24 AWG dual or 26 AWG triple.
Data 1 (D1) is used for Blade data signal, Data 2 (D2) is used for independent 5 on-board Hilt pcb led pixels data signal.
Or a "Sub-blade" configuration can be used to separate connector led pixels from main blade pixels for different blade styles, in this case only D1 data output is used and a V3 data input option for the NPXL connector (for more info look at page 4 of the NPXL manual).

Config file setup

```
#ifdef CONFIG_TOP
#include "proffieboard_v1_config.h"
#define NUM_BLADES 2
#define NUM_BUTTONS 2
#define VOLUME 1000
const unsigned int maxLedsPerStrip = 144;
#define CLASH_THRESHOLD_G 1.0
#define ENABLE_AUDIO
#define ENABLE_MOTION
#define ENABLE_WS2811
#define ENABLE_SD
#define SHARED_POWER_PINS
#endif

#ifdef CONFIG_PRESETS
Preset presets[] = {
  { "your sound font name", "tracks/your music track name.wav",
   your main blade style code,
   your connector blade style code, "your preset name"}
};

BladeConfig blades[] = {
{ 0,
   WS2811BladePtr<136, WS2811_ACTUALLY_800kHz | WS2811_GRB, bladePin, PowerPINS<bladePowerPin1, bladePowerPin2, bladePowerPin3> >(),     // Main Blade with 136 pixels
   WS2811BladePtr<5, WS2811_ACTUALLY_800kHz | WS2811_GRB, blade2Pin, PowerPINS<bladePowerPin1, bladePowerPin2, bladePowerPin3> >(),     // Connector "Hilt" pcb pixels
 CONFIGARRAY(presets) },
};
#endif

#ifdef CONFIG_BUTTONS
Button PowerButton(BUTTON_POWER, powerButtonPin, "pow");
Button AuxButton(BUTTON_AUX, auxPin, "aux");
#endif
```

define 2 blades (main blade style and connector pixels blade style)

make sure to add a *#define SHARED_POWER_PINS* line.
Also download and replace this file in "blades" folder
in ProffieOS root folder: **power_pin.h**

your main blade style code

your connector blade style code

hooked up to Data 1 output

PowerPINS are LED Negative outputs, they are same for main blade and connector pixels

hooked up to Data 2 output

MOUNTING OPTIONS

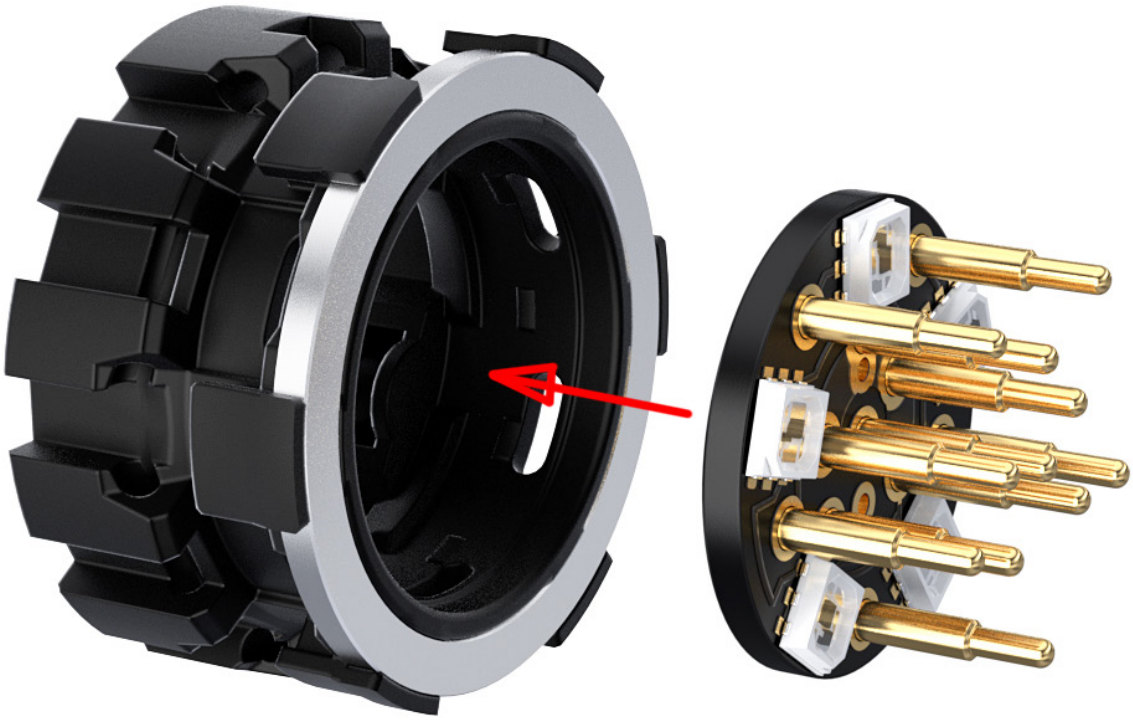
Hilt pcb basic 3D-printed Holder

For LONG pins:

– 7/8 inch blade holder
– 1 inch blade holder

For SHORT pins:

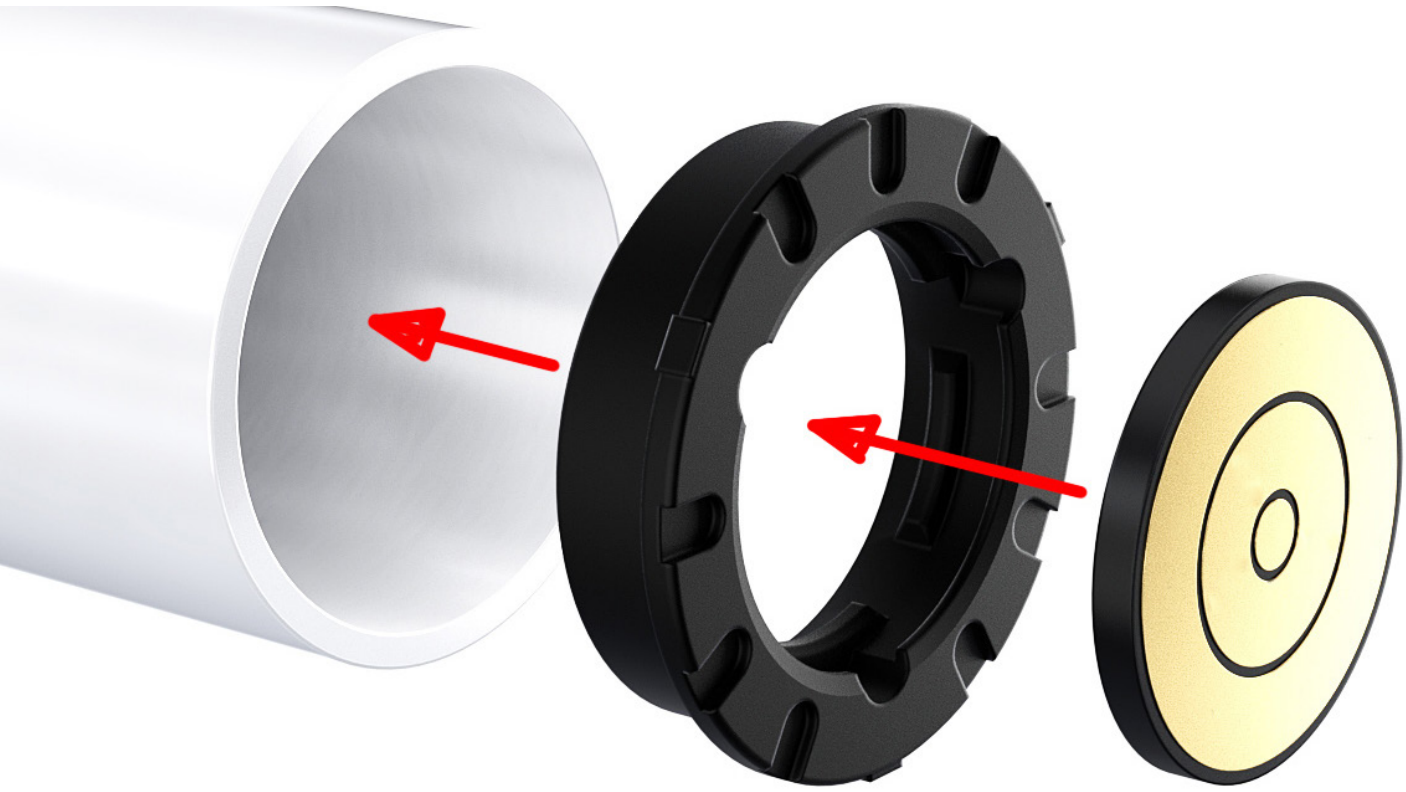
– 7/8 inch blade holder
– 1 inch blade holder
– Thin-Neck blade holder



4.2 MOUNTING OPTIONS

Blade pcb basic 3D-printed Holder

- 7/8 inch blade
- 1 inch Thin-walled blade
- 1 inch Thick-walled blade
- 1 inch Thin-walled blade for “Thin-Necks” V1
- 1 inch Thin-walled blade for “Thin-Necks” V2





CURRENT DRAW TEST

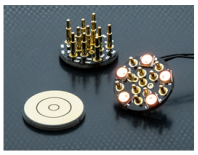
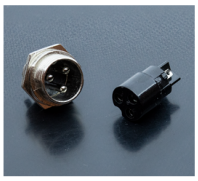
Neopixel WS2812B/SK6812 strips
tested approximate current consumption chart

Tested at 3.7V, 143 leds per strip, at max brightness

№ of strips	current	1 color without flicker / with flicker	2 colors mixed without flicker / with flicker	3 colors mixed for white without flicker / with flicker
1	Total	2 / 1.9 A	3.6 / 3.3 A	5.2 / 4.9 A
	Per LED	14 / 12.9 mA	12.6 / 11.5 mA	12.1 / 11.4 mA
2	Total	3.7 / 3.5 A	6.9 / 6.4 A	9.9 / 9.3 A
	Per LED	13 / 12.2 mA	12 / 11.1 mA	11.5 / 10.8 mA
3	Total	5.4 / 4.5 A	10.1 / 9.5 A	14.4 / 13.5 A
	Per LED	12.6 / 11.6 mA	11.8 / 11.1 mA	11.2 / 10.5 mA
4	Total	7.1 / 6.7 A	13 / 12.4 A	17.7 / 16.6 A
	Per LED	12.4 / 11.8 mA	11.4 / 10.8 mA	10.3 / 9.7 mA
5	Total	8.8 / 8.4 A	15.7 / 15 A	20.6 / 19.5 A
	Per LED	12.3 / 11.7 mA	11 / 10.5 mA	9.6 / 9.1 mA

MAX RATINGS TEST

Neopixel Connectors
tested continuous current max ratings chart

Brand/Name	Time	at 8A	at 10A	at 15A	at 20A
ShtokCustomWorx 	after 5 minutes	26 °C 100% efficiency **OK**	29 °C 100% efficiency **OK**	40 °C 100% efficiency **OK**	55 °C 98% efficiency **OK**
	after 15+ minutes	26 °C 100% efficiency **OK**	29 °C 100% efficiency **OK**	45 °C 100% efficiency **OK**	60+ °C 98% efficiency **OK**
6.3mm gold plated stereo jack 	after 5 minutes	50 °C 97% efficiency **OK**	60 °C 95% efficiency **NOT OK**	115 °C 56% efficiency connector melted **BAD**	
	after 15+ minutes	50 °C 97% efficiency **OK**	70+ °C 94% efficiency **NOT OK**		
GX16-3pin Aviation Plug 	after 5 minutes	28 °C 100% efficiency **OK**	30 °C 100% efficiency **OK**	40 °C 97% efficiency **OK**	55 °C 92% efficiency **NOT OK**
	after 15+ minutes	28 °C 100% efficiency **OK**	30 °C 100% efficiency **OK**	45 °C 97% efficiency **OK**	60+ °C <92% efficiency **NOT OK**

Which wire gauge is recommended to use for Positive and Negative power leads for maximum blade brightness efficiency

AWG gauge	Conductor Diameter Inches	Conductor Diameter mm	Conductor cross section in mm^2	Ohms per 1000 ft.	Ohms per km	Maximum amps for chassis wiring	
14	0.0641	1.62814	2.08	2.525	8.282	32	
15	0.0571	1.45034	1.65	3.184	10.44352	28	
16	0.0508	1.29032	1.31	4.016	13.17248	22	
17	0.0453	1.15062	1.04	5.064	16.60992	19	
18	0.0403	1.02362	0.823	6.385	20.9428	16	Neopixel strips Battery Recharge Port Kill Switch
19	0.0359	0.91186	0.653	8.051	26.40728	14	
20	0.032	0.8128	0.519	10.15	33.292	11	
21	0.0285	0.7239	0.412	12.8	41.984	9	
22	0.0253	0.64516	0.327	16.14	52.9392	7	
23	0.0226	0.57404	0.259	20.36	66.7808	4.7	
24	0.0201	0.51054	0.205	25.67	84.1976	3.5	Tri-Cree LED Battery Recharge Port Kill Switch
25	0.0179	0.45466	0.162	32.37	106.1736	2.7	
26	0.0159	0.40386	0.128	40.81	133.8568	2.2	
27	0.0142	0.36068	0.102	51.47	168.8216	1.7	
28	0.0126	0.32004	0.080	64.9	212.872	1.4	
29	0.0113	0.28702	0.0647	81.83	268.4024	1.2	
30	0.01	0.254	0.0507	103.2	338.496	0.86	Everything else
31	0.0089	0.22606	0.0401	130.1	426.728	0.7	
32	0.008	0.2032	0.0324	164.1	538.248	0.53	

Chart from PowerStream.com

Neopixel strips build (3-17 amperes load)			Tri-Cree LED build (1-4 amperes load)
2-strip	3-strip	4-strip	28-24 AWG recommended for battery wiring, choose regarding particular build *30 AWG possible for single 3W Cree LED wiring (one wire per die)*
22 AWG single or 24 AWG dual in parallel	20 AWG single or 23 AWG dual in parallel	18 AWG single or 22 AWG dual in parallel	

For all other components except Neopixel blade strips, high power Tri-Cree LEDs, battery and recharge port/Kill Switch – a 30-32 AWG wire can be used because they are low current circuits (5-500mA) (accent leds, activation and AUX switches, speaker, bluetooth module, RICE port etc.).